Team Project

Deadline for submission: Last Day of the course
Late submissions will not be accepted.

Version 3 – MovinOn Co.

Student Name(s)	Student ID

You will work in a team, with **3**-5 **students per team**, for the term project database system. Students should form their own teams and send an email to the TA with the names of the team members, if the TA has not assigned you to any teams.

You can choose MySQL Server or Microsoft SQL Server to develop the project.

Case Study – The following exercise allows you to design a database

In Assignment 1, you created the database design for MovinOn Inc. The database will be used to manage data about the company and its operations. David Bowers, the company's president, and Robert Iko, the company's information systems specialist, reviewed your design and worked to finalize it. The Figure 2.1 shows the database design that David and Robert approved.

David and Robert are satisfied that the design in Figure 2.1 will output the data needed to manage the business. With the design approved, you can begin developing the database.

1. Use your own design from Assignment 1 or compare the database design you developed with the one shown in Figure 2.1. If necessary, change your database and table designs to match the ones shown in the figure. If you determine that you need to add or change fields in your table designs, be certain to carefully consider and then set the properties that will support the data being stored in those fields.

2. Create a new database named **MovinOn_TeamNo**.

3. Use the data model shown in Figure 2.1 as a guide to help you develop the database. Be certain to specify the field names, data types, field descriptions, and field sizes as you create each table. Create validation rules as necessary to ensure that users enter consistent, complete, and accurate data in the tables. During the discovery phase (database analysis and logical design), David and Robert gave you some important information that you must consider in your table designs.

4. Check your database carefully to ensure that you have created all the tables, that you have created all the fields and set their properties.

5. Check all your tables for accuracy and ensure that all your relationships have been established according to the design shown in Figure 2.1.

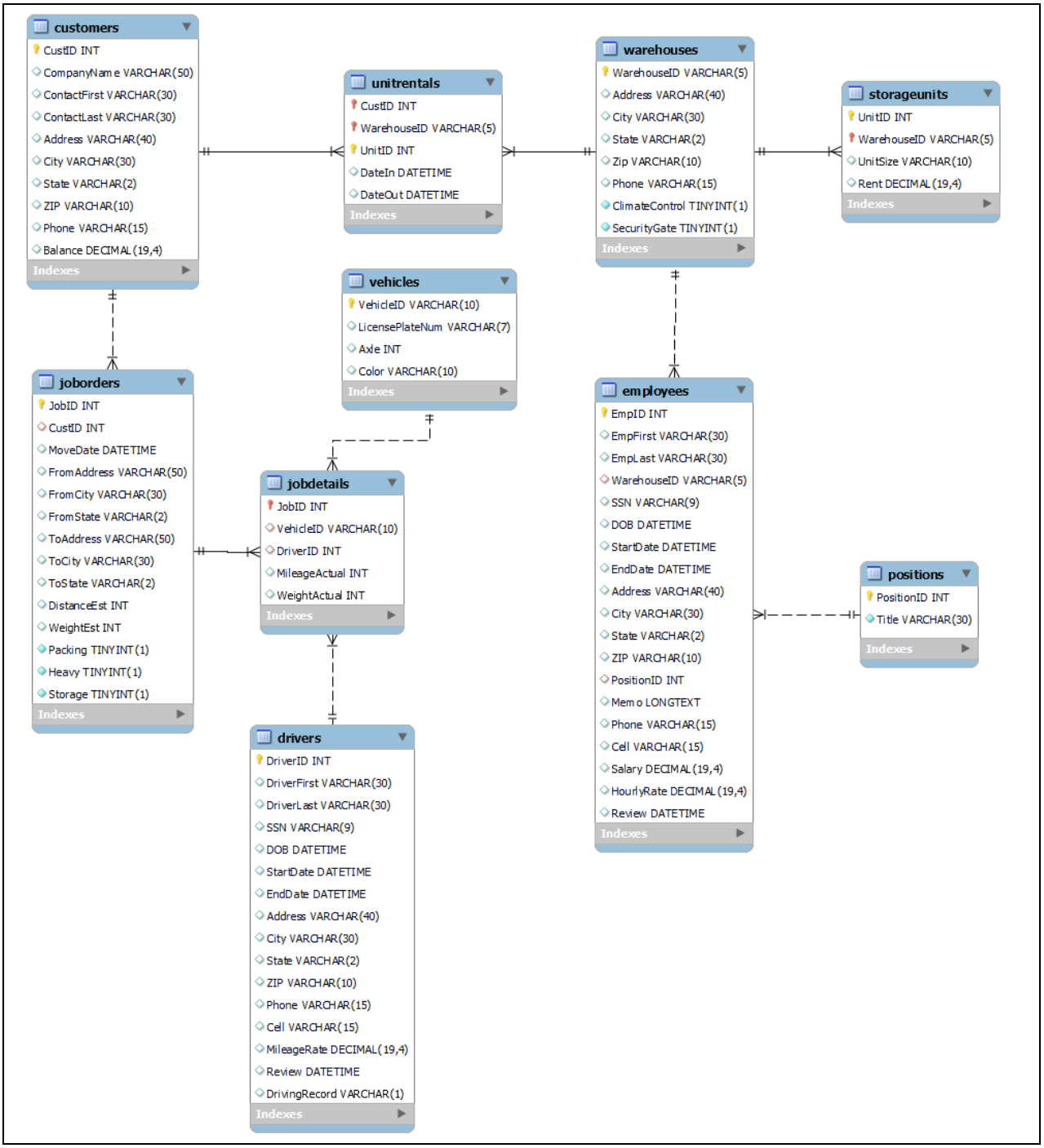

Figure 2.1: MovinOn Inc. Database Design

6. Add records to each table.
   In some cases, you might need to import data stored in an Excel workbook into a table. If you have not created the table in the database, you can import the data and create the table at the same time. Robert received some Excel files from David containing data that he needs to store in the

database. One of those files, Employee.xlsx (located in the Assignment 2 Data folder), contains data about the employees. Robert decides to create **Employee** table by importing the data from the Employee.xlsx Excel file.

7. Now that you have worked with Robert Iko at the MovinOn moving and storage company to develop the design for the MovinOn database, he explains that their most pressing task is to serve the needs of the human resources department. Darnell Colmenero is an administrative assistant responsible for many human resources tasks and asks for your help extracting information from the MovinOn database. Although an outside company processes payroll for MovinOn, Darnell and others maintain complete employment information and strive to meet management's goal of recruiting and retaining skilled, qualified employees who are well trained in customer service. Having employees working in three warehouses in three states has made it difficult to track employee information, and the potential merger and expansion means that human resources must take advantage of the MovinOn database to maintain and retrieve employee information. Darnell asks for your help in filtering data and creating queries that provide the information that he needs.

Enhancing Data Entry and Verifying Data for MovinOn Inc.

In addition to tasks completed in Assignment 01, This project covers the advanced topics of a databased design and implementation (view, function, stored procedure, trigger, and transaction objects).

David Bowers, general manager of MovinOn, met with his warehouse managers to discuss the progress you have made in developing the MovinOn database. Because MovinOn Inc. recently adopted a pension plan for all employees, the managers want to inform employees when they are fully vested in the plan, which happens after five years of service.

David asks if you can add a message (script) that indicates when an employee is fully vested based on the length of service. He also wants you to calculate an employee's years of service. For active employees, he wants you to hide the **End Date** field; for inactive employees, he wants you to hide the **Review field**. Finally, to help with data entry, the warehouse managers want to verify phone and cell phone area codes entered into Employee Data.

Complete the following:

- Add a new script that uses a calculation to accurately determine the years of service for each active and inactive employee.

A simple algorithm to determine whether an employee is vested: If the employee end date is null (the employee is not terminated yet) and the employee has a started (hired) date, then calculate the employee's years of service. Otherwise, the employee is not vested.

- Verify telephone area codes for employee phone numbers. For employees living in Oregon, the valid area codes are 541, 503, and 971; for employees living in Washington, the valid area codes are 425, 360, 206, 509, and 253; for employees living in Wyoming, the valid area code is 307. Warehouse managers should be able to enter phone area codes for other states not included in the lists.

A Simple algorithm: Check if the employee's years of service are greater than five years, then this employee is eligible, otherwise the employee is not vested yet. To Verify the telephone area codes for Oregon, Washington, and Wyoming, you can check the first three digits.

Orlando area code (OR): 541, 503, or 971. Otherwise print the message "OR phone area codes should be 541, 503, or 971".

Washington area code (WA): 425, 360, 206, 509, or 253. Otherwise print the message "WA phone area codes should be 425, 360, 206, 509, or 253".

Wyoming area code (WY): 307. Otherwise print the message "WY phone area codes should be 307".

For active employees, hide the End Date. For inactive employees, display the End Date.

- Create a script that returns the earnings of each employee. Maria wants to view each employee's annual salary. For salaried employees, the Salary field value represents the annual salary. However, for active, hourly paid employees, the field value displays the hourly rate but not an annual salary. For these employees, Don needs to multiply the hourly rate by 2,080, which is the product of 40 hours per week and 52 weeks per year.

- Don needs to calculate the number of years between an employee's start date and today's date to determine the number of years of service. For the employee age request, Don needs to calculate the number of years between an employee's date of birth and today's date to determine the employee's age.

In this project, you work with Kristina Romano, an accountant in the Washington warehouse of MovinOn Inc. Kristina needs to prepare a series of financial reports to present to David Bowers, owner of MovinOn, the warehouse managers, and the accountants at MovinOn at their annual meeting. They are naturally interested in how much revenue recent jobs have generated, as well as the labor costs that offset this revenue.

Kristina suggests that you use the following guidelines as you design the reports:

- Provide an appropriate title that clearly states the purpose of each report.
- Show all dollar amounts with a dollar sign and two decimal places.

- Before the annual meeting, Kristina plans to mail copies of some reports to the warehouse managers. Because she will send materials to each manager at different times, she needs a way to indicate which warehouse she wants before she prints the mailing label. The label she produces should include the manager's warehouse number next to their name, along with other appropriate mailing information. She wants you to name this report **WarehouseManagerReportLabels.**

The next report that Kristina needs should show the income from recent moving jobs. MovinOn charges $.70 per mile plus $.20 per pound for each job. From this total, Kristina deducts the driver's payment to determine the net income for the moving job. Drivers receive $50 for each job plus their mileage rate (that is, the driver's rate multiplied by the number of miles plus $50).

- Kristina asks you to create an income report that provides the job ID, the date of the move, the driver's name and rate, the mileage and weight of the job, and the income calculations (The income could be calculated as Mileage Actual * 0.7 plus Weight Actual * 0.2 from table Job Detail.) The Net Income = Income – Driver Payment. Kristina wants to show the total income, the total payments to drivers, and the total net income. (*Hint*: Consider creating a query as the basis for this report.) Name the report **JobRevenueReport**.

- Kristina also needs a report that shows the income from the storage units. Group the information by warehouse and show the name of the renter so that it's easy to identify the renter by last name. The report should also include the rent per unit, the total rent for each warehouse, and a grand total of rent for all warehouses. Name the report **StorageRevenueReport**.

- Kristina suspects that MovinOn could increase its income from storage units by encouraging more long-term rentals. Add a calculation to StorageRevenueReport that shows how long each renter has rented a storage unit. Show the figure in years with one decimal place.

Additional questions that are frequently asked include the following:

1. How many storage units did the company rent last year?

2. What percentage of customers have rented at least one unit?
3. What was the greatest number of rents by an individual?
4. What is the average length of a rental period?
5. What are the company's peak rental months?

As per team effort, each team member must ask at least one question regarding the moving and create the corresponding script.

The answers to these questions can be created as database objects: views, functions, stored procedures, or triggers. What to design for each question is up to you, but you must cover all objects.